# Semantic Branching Cheat Sheet

Semantic Entropy Clustering for Diverse Agentic Code Generation

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**Compute Budget**

Both baseline and branching are allowed **250 steps per trajectory**. The baseline runs a single greedy trajectory (250 steps total). Branching spawns up to $B$=30 trajectories, each with the same 250-step cap. Results: baseline 5/10, branching 4/10, **union 7/10** — the methods are complementary.

## 1 NLI Relevance Scoring (Search Phase)

During the search phase the agent runs bash commands (`grep`, `find`, `cat`) to explore the repo. After *each* step we score how relevant the finding is to the bug.

### 1.1 Why Not Pure DeBERTa NLI?

DeBERTa NLI is trained on **natural language** sentence pairs, not code-to-bug pairs.

| Problem | Result |
|---|---|
| No shared context between code output and bug description | Entailment ≈ 0 always |
| Too much shared context (copy-pasted keywords) | Entailment ≈ 1 always |

**Solution**: use the LLM itself to bridge the domain gap via two cheap calls.

### 1.2 Step 1 — LLM Summarization

Converts raw search output into a natural language sentence.

**Summarization Prompt**

An AI agent is debugging a software issue. It ran a search command and got this result. In one sentence, what did this search find and why might it be relevant?

Agent's reasoning: `{thought}` ← 300 chars
Search result (truncated): `{observation}` ← 500 chars
One-sentence summary of what was found:

`temperature=0.0`, `max_tokens=80`

**Output example**: *"Found the Permutation constructor in permutations.py which handles cycle notation input."*

### 1.3 Step 2 — LLM Relevance Rating

**Relevance Prompt**

Rate how relevant this investigation finding is to the bug below.
Bug: `{problem_statement}` ← 300 chars
Finding: `{finding_summary}` ← from Step 1
Rate relevance from 0 (completely irrelevant) to 10 (directly identifies the bug).
Respond with ONLY a single number 0–10.

`temperature=0.0`, `max_tokens=5`

**Post-processing**: regex extracts first integer → clamp to $[0, 10]$ → divide by 10 → score $\in [0, 1]$.

### 1.4 Why Not DeBERTa NLI for Relevance?

We tested using DeBERTa entailment instead of LLM-as-judge (even with the LLM summary bridge). Results on sympy-16766:

| Summary | NLI | LLM |
|---|---|---|
| "Found code generation utilities" | 0.003 | 0.7 |
| "Found printing functionality files" | 0.038 | 0.7 |
| "Found Python code printers module" | 0.618 | 0.7 |
| "Found Indexed class docs" | 0.005 | 0.7 |
| "mpmath missing, import failed" | 0.017 | 0.3 |

**Why?** NLI measures *logical entailment* ("does A imply B?"), not *topical relevance*. "Found Indexed class" doesn't logically imply "Indexed printing is broken" — but it's clearly relevant. NLI works for **clustering** (same-structure intent comparisons) but not for relevance scoring.

### 1.5 Saturation Detection

**Saturation Rule**

$$\text{consec\_low} \geq N_{\text{cap}} \implies \textbf{Strategy Phase}$$

A step is "low relevance" if score $\leq \theta$ (default $\theta = 0.5$).

## 2 Generating 5 Diverse Strategies

After search saturation: (1) build a **search report**, then (2) prompt for strategies.

### 2.1 Search Report Prompt

**Search Report Prompt**

An AI coding agent investigated a software bug. Below is its exploration history with the actual code it found.
Write a structured report with these sections:
1. ROOT CAUSE: What is the bug and why?
2. RELEVANT CODE: Verbatim snippets with file/function/line.
3. FIX POINTS: Different places a fix could be applied.
IMPORTANT: Include actual code from observations. Do NOT paraphrase.
Exploration history: `{history}` ← last 15 relevant steps, 8k chars

`temperature=0.0`, `max_tokens=1500`

### 2.2 Strategy Proposal Prompt (Verbatim)

**Strategy Proposal Prompt**

A software bug needs to be fixed. Here is the problem description and investigation findings.
**## Problem**
`{problem_statement}` ← 2000 chars
**## Investigation Findings**
`{search_report}` ← 4000 chars
**## Task**
Propose exactly `{n}` FUNDAMENTALLY DIFFERENT code-level strategies to fix this bug.
CRITICAL RULES:
- Each strategy MUST take a COMPLETELY DIFFERENT APPROACH
- Each strategy MUST modify DIFFERENT lines, functions, or files
- Each strategy MUST produce a structurally different patch
- Do NOT propose strategies that all change the same line
- Be SPECIFIC: name exact function, file, and line range
- Describe the actual code transformation

Think about fundamentally different approaches:
- Fix at different levels of the call stack
- Fix in different files or modules
- Use different mechanisms (validation vs normalization vs delegation)
- Address different root causes

IMPORTANT: Before writing each strategy, explicitly state what makes it FUNDAMENTALLY DIFFERENT from all previous strategies.
Format: `STRATEGY 1: [file/function] -- [specific code change]`

$n = 5$, `temperature=1.0` (high for diversity), `max_tokens=1200`

### 2.3 Iterative Rejection

If $< 5$ strategies parsed, a **second pass** explicitly rejects already-seen approaches:

**Rejection Clause**

Already Proposed Strategies (DO NOT repeat):
– ALREADY PROPOSED: `{strategy_1}`
– ALREADY PROPOSED: `{strategy_2}` ...
Propose `{n_more}` NEW strategies targeting DIFFERENT code locations using DIFFERENT mechanisms.

## 3 Bidirectional Entailment, Clustering & Semantic Entropy

### 3.1 DeBERTa NLI Classification

**Model**: DeBERTa-large fine-tuned on MNLI.

**Input**: (premise, hypothesis) pair → tokenized as:

`[CLS]` premise `[SEP][SEP]` hypothesis `[SEP]`

**Forward pass**: DeBERTa → logits $\boldsymbol{l} = (l_0, l_1, l_2) \in \mathbb{R}^3$

**Softmax to Probabilities**

$$p_k = \frac{\exp(l_k)}{\sum_{j=0}^{2} \exp(l_j)} \qquad k \in \{0, 1, 2\}$$

**Output**: $\{p_0, p_1, p_2\}$ where $p_0 + p_1 + p_2 = 1$

| Index | Label | Meaning |
|---|---|---|
| 0 | contradiction | Premise contradicts hypothesis |
| 1 | neutral | Neither entails nor contradicts |
| 2 | entailment | Premise implies hypothesis |

### 3.2 Entailment Check

Given threshold $\theta$ (default $= 0.5$):

**One-Way Entailment**

$$\text{ENT}(A, B) = \left[\, p_2(A,\ B) > \theta \,\right]$$

where $p_2(A, B) = P(\text{entailment} \mid \text{prem}{=}A,\ \text{hyp}{=}B)$

### 3.3 Bidirectional Entailment

Two texts are **semantically equivalent** iff each entails the other:

**Bidirectional Entailment**

$$\text{BIDIR}(A, B) = \text{ENT}(A, B) \ \wedge\ \text{ENT}(B, A)$$

**Why bidirectional?**

| Pair | A→B | B→A | Bidir |
|---|---|---|---|
| "Fix constructor" vs "Modify code" | ✓ | × | **No** |
| "Remove dup check" vs "Delete redundant val." | ✓ | ✓ | **Yes** |
| "Add normalization" vs "Fix constructor" | × | × | **No** |

### 3.4 Clustering Algorithm

**Algorithm 1** from Farquhar et al. 2024 / Kuhn et al. 2023.

**Context concatenation** (critical step from the paper):

Before any NLI comparison, prepend the problem context:

$$\text{nli\_input}_i = \text{context} + \text{intent}_i$$

**Greedy Sequential Clustering**

**In**: intents $[s_0, \ldots, s_{N-1}]$, context $C$, thr. $\theta$
**Out**: clusters $\mathcal{K} = \{C_1, \ldots, C_K\}$

$\mathcal{K} \leftarrow \emptyset$
**for** $i = 0$ **to** $N{-}1$:
  assigned $\leftarrow$ False
  **for each** cluster $C_k \in \mathcal{K}$:
    $r \leftarrow C_k$.rep
    fwd $\leftarrow p_2(C{+}s_r,\ C{+}s_i)$
    bwd $\leftarrow p_2(C{+}s_i,\ C{+}s_r)$
    **if** fwd $> \theta$ **and** bwd $> \theta$:
      add $i$ to $C_k$; assigned; **break**
  **if not** assigned:
    new cluster with rep $= i$

| Property | Detail |
|---|---|
| Order-dep. | First intent → cluster rep |
| Greedy | Assigns to *first* match |
| Single rep | Compares to cluster's first member |
| Complexity | $O(N{\times}K)$ NLI calls, $K \leq N$ |

### 3.5 Semantic Entropy

**Semantic Entropy (Discrete Variant)**

$$H = -\sum_{k=1}^{K} p(c_k) \cdot \ln\big(p(c_k)\big)$$

$$\text{where} \quad p(c_k) = \frac{|c_k|}{N}$$

**Every variable**:

| Sym. | Meaning |
|---|---|
| $H$ | Semantic entropy — diversity in nats |
| $K$ | Number of semantic clusters |
| $k$ | Cluster index, 1 to $K$ |
| $c_k$ | Cluster $k$ (set of candidate indices) |
| $\lvert c_k \rvert$ | Size of cluster $k$ |
| $N$ | Total candidates ($\sum_k \lvert c_k \rvert$) |
| $p(c_k)$ | Cluster probability $= \lvert c_k \rvert / N$ |
| ln | Natural log (base-$e$) |

**Bounds**:

| Scenario | Sizes | $H$ |
|---|---|---|
| All in one cluster | [5] | 0 |
| Two equal groups | [3, 2] | 0.673 |
| All singletons | [1, 1, 1, 1, 1] | $\ln 5 \approx 1.609$ |

**Min**: $K{=}1 \Rightarrow p{=}1 \Rightarrow -1 \cdot \ln 1 = 0$
**Max**: $K$ equal clusters $\Rightarrow H = \ln K$

### 3.6 Branching Decision

**Branch Criterion**

$$\text{branch} = \left[\, H > \tau \,\right] \qquad \text{default } \tau = 0.5$$

- $H > \tau$: diverse → **branch** — each cluster spawns a trajectory
- $H \leq \tau$: converged → **no branch** — take greedy action

### 3.7 Worked Example

**5 strategies**, clustered with $\theta = 0.5$:

| | Strategy |
|---|---|
| $s_0$ | Normalize overlapping cycles in constructor |
| $s_1$ | Raise ValueError when cycles share elements |
| $s_2$ | Canonicalize cycle input in from_sequence |
| $s_3$ | Add validation in Permutation constructor |
| $s_4$ | Fix _af_new helper for overlapping cycles |

**Clustering trace**:

| $i$ | fwd | bwd | Result |
|---|---|---|---|
| 0 | – | – | New cluster $A$={0} |
| 1 | .12 | – | $< \theta \rightarrow$ new $B$={1} |
| 2 | .78 | .65 | Both $> \theta \rightarrow A$={0, 2} |
| 3 | .72 | .58 | Both $> \theta \rightarrow A$={0, 2, 3} |
| 4 | .21 | – | $< \theta$ for $s_0$, .15 for $s_1 \rightarrow$ new $C$={4} |

**Result**: 3 clusters, sizes $[3, 1, 1]$

**Entropy**:

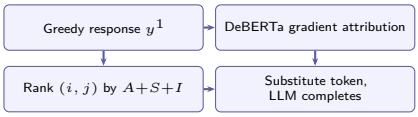
$$p(A) = \tfrac{3}{5} = 0.6, \quad p(B) = p(C) = \tfrac{1}{5} = 0.2$$

$$\begin{aligned} H &= -\big[0.6\ln(0.6) + 0.2\ln(0.2) + 0.2\ln(0.2)\big] \\ &= -\big[0.6(-0.511) + 0.2(-1.609) + 0.2(-1.609)\big] \\ &= -(-0.306 - 0.322 - 0.322) \\ &= \mathbf{0.950} \end{aligned}$$

**Decision**: $0.950 > 0.5 = \tau \implies$ **BRANCH** into 3 trajectories.

## 4 SDLG — Semantically Diverse Language Generation

Aichberger et al. (2025). Generates diverse alternatives by gradient-based token attribution through DeBERTa, then substitution + LLM re-completion.




### 4.1 Step 0: Greedy Response

Generate baseline at temp = 0. Split into:

| Part | Contents |
|---|---|
| THOUGHT | Natural language reasoning |
| CODE | Bash command in `"‘mswea_bash_command"‘` |

### 4.2 Step 1: Attribution Score $A_i$

**Goal**: Which tokens carry the most semantic weight?

#### 4.2.1 Setup — Self-Entailment

Feed text as **both** premise and hypothesis:

$$\texttt{[CLS]}\ \underbrace{\text{text}}_{\text{prem.}}\ \texttt{[SEP]}^2\,\texttt{[SEP]}^2\,\texttt{[SEP]}^2\,\texttt{[SEP]}^2\ \underbrace{\texttt{text}}_{\text{hyp.}}\ \texttt{[SEP]}$$

#### 4.2.2 Loss — Target Contradiction

**Contradiction Loss**

$$\mathcal{L} = \mathrm{CE}(\boldsymbol{l},\ \text{target}{=}0) = -\log \frac{\exp(l_0)}{\sum_k \exp(l_k)}$$

**Why contradiction?** The text trivially entails itself. The gradient of the contradiction loss reveals which tokens, if changed, would flip self-entailment to self-contradiction — i.e., the tokens that carry the meaning.

#### 4.2.3 Backprop & Attribution

Backpropagate $\mathcal{L}$ through the embedding layer to get $\nabla_{\boldsymbol{z}_i}\mathcal{L}$ for each token $i$. Extract **hypothesis portion only**.

**Attribution Score**

$$A_i \;=\; \big\| \boldsymbol{z}_i \odot \nabla_{\boldsymbol{z}_i}\mathcal{L} \big\|_2$$

**Every symbol**:

| Sym. | Meaning | Shape |
|---|---|---|
| $i$ | Token position in hypothesis | int |
| $\boldsymbol{z}_i$ | Embedding vector of token $i$ | $[D]$ |
| $D$ | Embedding dim (1024 for DeBERTa-large) | int |
| $\mathcal{L}$ | Contradiction loss | scalar |
| $\nabla_{\boldsymbol{z}_i}\mathcal{L}$ | Gradient of $\mathcal{L}$ w.r.t. $\boldsymbol{z}_i$ | $[D]$ |
| $\odot$ | Hadamard (element-wise) product | $[D]$ |
| $\|\cdot\|_2$ | L2 norm: $\sqrt{\sum_d (\cdot)^2}$ | scalar |
| $A_i$ | Attribution score for token $i$ | $\geq 0$ |

**Intuition for each piece**:

| Component | What it captures |
|---|---|
| $\boldsymbol{z}_i$ | Large embedding = strong signal |
| $\nabla_{\boldsymbol{z}_i}\mathcal{L}$ | Large gradient = loss sensitive to this token |
| $\boldsymbol{z}_i \odot \nabla$ | Both conditions simultaneously |
| $\|\cdot\|_2$ | Collapses $D$ dims to one scalar |

**Normalization**: $A_i \leftarrow A_i \,/\, \max_i(A_i)$ so $A_i \in [0, 1]$.

**Word boundary filter**: Only word-initial BPE tokens (prefix Ġ or _) are candidates. Avoids sub-word fragments.

### 4.3 Step 2: Substitution Score $S_{ij}$

**Goal**: For position $i$, which replacement $j$ from the vocabulary shifts meaning most along the gradient?

**Substitution Score**

$$S_{ij} \;=\; \frac{(\boldsymbol{z}_i - \boldsymbol{z}_j)\ \cdot\ \nabla_{\boldsymbol{z}_i}\mathcal{L}}{\|\boldsymbol{z}_i - \boldsymbol{z}_j\|_2\ \cdot\ \|\nabla_{\boldsymbol{z}_i}\mathcal{L}\|_2}$$

**Every symbol**:

| Sym. | Meaning |
|---|---|
| $\boldsymbol{z}_i$ | Embedding of original token at pos $i$ |
| $\boldsymbol{z}_j$ | Embedding of candidate replacement $j$ |
| $\boldsymbol{z}_i - \boldsymbol{z}_j$ | How much the embedding changes |
| $\nabla\mathcal{L}$ | Direction increasing contradiction |
| $\cdot$ | Dot product |
| $S_{ij}$ | Cosine sim. of change vs gradient |

**Interpretation**:

| $S_{ij}$ | Meaning |
|---|---|
| $\approx +1$ | Moves along gradient → max meaning change |
| $\approx 0$ | Orthogonal → no semantic shift |
| $\approx -1$ | Against gradient → reinforces original |

Computed **server-side** over the full vocabulary ($V \approx$ 128k):

```
diff = z_i.unsqueeze(0) - emb_matrix # [V, D]
diff_norms = diff.norm(dim=-1) # [V]
s_ij = (diff @ grad_i) / (diff_norms * grad_norm)
```

Top-$K$ ($K$=20) replacements selected. Self-substitution excluded.

### 4.4 Step 3: Importance Score $I_{ij}$

**Goal**: Is the substitute plausible under the LLM?

**Importance Score**

$$I_{ij} \;=\; p\big(v_j \,\big|\, y_{<i},\ x,\ w\big)$$

| Symbol | Meaning |
|---|---|
| $v_j$ | Candidate replacement token |
| $y_{<i}$ | All text before position $i$ |
| $x$ | Full conversation context (history) |
| $w$ | LLM weights (Qwen3-Coder-30B) |
| $p(\cdot)$ | LLM's next-token distribution |

**Computed via vLLM**: send prefix to `/v1/completions` with `logprobs=20`, convert: $I_{ij} = \exp(\text{logprob}_j)$.

**Why $I_{ij}$ matters**

Without $I_{ij}$, SDLG might substitute "remove" → "xylophone" (high $S_{ij}$ because embeddings are far apart, but nonsensical). $I_{ij}$ filters for tokens the LLM would actually generate.

### 4.5 Step 4: Combined Ranking

**Combined Score**

$$\text{Combined}_{ij} = \frac{A_i + S_{ij} + I_{ij}}{3}$$

| | Range | Measures | Source |
|---|---|---|---|
| $A_i$ | $[0, 1]$ | Position importance | DeBERTa $\nabla$ |
| $S_{ij}$ | $[-1, 1]$ | Meaning shift | emb+$\nabla$ |
| $I_{ij}$ | $[0, 1]$ | LLM plausibility | vLLM |

All $(i, j)$ pairs sorted by $\text{Combined}_{ij}$ descending.

### 4.6 Step 5: Prefix Truncation & Completion

#### 4.6.1 THOUGHT-Level (Strategic Diversity)

Changes the *reasoning*, causing a different fix approach:

**Thought Substitution**

1. Find original token in thought text
2. Replace with substitute
3. **Truncate everything after** substitution point
4. Send truncated thought as `assistant` message prefix
5. LLM regenerates rest of reasoning **AND** code

**Example**:

- Greedy: "I should **remove** the duplicate check…"
- Substitute: "remove" → "compose"
- Prefix: "I should compose"
- LLM completes: "…the cycles before validation…" + *new code*

#### 4.6.2 CODE-Level (Tactical Diversity)

Keeps reasoning fixed, diversifies the implementation:

**Code Substitution**

1. Keep thought text **unchanged**
2. Find original token in code block
3. Replace with substitute, truncate
4. LLM completes the code from there
5. Result: same reasoning, different patch

#### 4.6.3 Budget Split

For $N$ total candidates (default $N$=5):

| Source | Count |
|---|---|
| Greedy response (always kept) | 1 |
| THOUGHT-level SDLG | $\lceil (N-1)/2 \rceil = 2$ |
| CODE-level SDLG | $\lfloor (N-1)/2 \rfloor = 2$ |

### 4.7 Fallback

If SDLG produces zero alternatives → **temperature sampling** at $T$=0.7.